

MediScan AI

Multi-Modal Medical Imaging Diagnostic Assistant

Overview

MediScan AI is a multi-modal deep learning system that screens four of the most common and clinically urgent diagnostic scenarios in radiology — brain tumors, pneumonia, COVID-19, and kidney disease (cysts, stones, and tumors) — from a single web interface. Four separate convolutional neural networks, each built on an EfficientNet backbone pretrained on ImageNet, were trained using a two-phase transfer learning strategy (frozen head-training followed by careful low-learning-rate fine-tuning) to classify Brain MRI, Chest X-Ray, Chest CT, and Kidney CT images respectively.

The four models are unified into a single production-ready application built with Gradio and deployed on Hugging Face Spaces. Users upload or photograph a scan, provide basic patient details, and receive an AI-generated diagnosis with a confidence score, a color-coded severity rating, and plain-language guidance on recommended next steps. A built-in AI chatbot, powered by Groq's llama-3.3-70b-versatile, is automatically aware of the user's latest scan result and can answer follow-up questions about the diagnosed condition — while staying strictly within its supported medical topics.

The kidney disease model additionally introduces Grad-CAM–based visual explainability: a gradient-weighted class activation map is overlaid on a CLAHE-contrast-enhanced version of the input CT scan, highlighting the image regions that most influenced the model's prediction. This explainability layer is currently implemented for the kidney model, with extension to the other three models identified as a planned next step.

Across the four tasks, the models reached 95.6% accuracy / 0.99 AUC (brain tumor, 4-class), 91.3% accuracy / 0.97 AUC (COVID-19 CT), 89–92% accuracy / 0.95 AUC (pneumonia X-ray), and 97.9% accuracy / 0.999 AUC (kidney disease, 4-class), with special attention paid throughout to recall — the metric that matters most when a missed diagnosis carries real clinical risk.

Key Features

- Four independently trained, disease-specific deep learning classifiers unified in one app
- Two-phase transfer learning (frozen backbone → targeted fine-tuning) for each model
- Confidence-aware UI with severity color-coding, low-confidence warnings, and a probability-breakdown donut chart
- Grad-CAM + CLAHE visual explainability for the kidney disease model, showing which scan regions drove the prediction
- Context-aware AI chatbot that references the user's most recent diagnosis
- Session-based patient scan history log
- Upload, webcam capture, or clipboard-paste image input; light/dark theme toggle
- Clear medical disclaimers throughout — built as an educational/research screening tool, not a diagnostic device

Technologies Used

Python · TensorFlow / Keras 3 · EfficientNetB0 / B3 / B4 · scikit-learn · NumPy · Pandas · Matplotlib · Seaborn · OpenCV (Grad-CAM/CLAHE visualization) · split-folders · Gradio · Groq API (Llama 3.3 70B) · Kaggle / kagglehub · Google Colab · Hugging Face Spaces

Machine Learning Models

Model	Backbone	Task	Input Size
Brain Tumor Classifier	EfficientNetB0	4-class MRI classification (glioma, meningioma, pituitary tumor, no tumor)	224×224

COVID-19 CT Classifier	EfficientNetB4	Binary CT classification (COVID vs. non-COVID)	380×380
Pneumonia X-Ray Classifier	EfficientNetB3	Binary X-ray classification (pneumonia vs. normal)	384×384
Kidney Disease Classifier	EfficientNetB0 (B3 configurable)	4-class CT classification (cyst, normal, stone, tumor)	224×224 (300×300 on B3)

Dataset

- Brain Tumor MRI Dataset (Kaggle) — 7,200 balanced MRI images across 4 classes
- SARS-CoV-2 CT-Scan Dataset (Kaggle) — 2,481 chest CT images, COVID vs. non-COVID
- Chest X-Ray Images (Pneumonia) Dataset (Kaggle, Paul Mooney) — 6,464 chest X-ray images, pneumonia vs. normal
- CT KIDNEY DATASET: Normal-Cyst-Tumor-and-Stone (Kaggle) — 12,446 kidney CT images across 4 classes (Cyst: 3,709 · Normal: 5,077 · Stone: 1,377 · Tumor: 2,283)

Results Summary

Model	Accuracy	ROC-AUC	F1-Score (macro)
Brain Tumor (EfficientNetB0)	95.6%	0.99	0.95
COVID-19 CT (EfficientNetB4)	91.3%	0.97	0.91
Pneumonia X-Ray (EfficientNetB3)	89–92%	0.95–0.96	0.89–0.93
Kidney Disease (EfficientNetB0)	97.9%	0.999	0.972

Skills Demonstrated

- Transfer learning and staged fine-tuning strategy design for CNNs on medical imaging data
- Handling class imbalance with computed class weights, and mitigating catastrophic forgetting during fine-tuning
- Rigorous model evaluation: confusion matrices, per-class precision/recall/F1, ROC-AUC, and custom composite scoring
- Targeted error analysis and iterative model improvement (identifying and fixing specific false-negative patterns)
- Model explainability with Grad-CAM: constructing a gradient-tape sub-model around a nested Keras 3 functional backbone to visualize class-discriminative regions, combined with CLAHE contrast enhancement for clinical readability
- Reproducible dataset partitioning with stratified per-class train/validation/test splitting (split-folders)
- End-to-end ML product development: from raw Kaggle datasets to a deployed, multi-model, LLM-augmented web application
- Debugging real production and training issues (probability/logits normalization, class-index ambiguity, EfficientNet preprocessing conventions across Keras versions)
- Full-stack AI app development with Gradio, including custom UI/UX, theming, and an integrated conversational assistant